



ORDNANCE SURVEY GB

OS OPEN GREENSPACE – TECHNICAL SPECIFICATION

Version history

Version	Date	Description
1.0	03/2017	Initial release
1.1	01/2019	Introduction of GeoPackage
1.2	04/2021	Introduction of vector tiles

Purpose of this document

This is the Technical Specification for the OS Open Greenspace product. This Specification provides greater insight into this product and its potential applications. For information on the contents and structure of OS Open Greenspace, please refer to the Product Guide and Getting Started Guide.

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I. Introduction

I.1 Product overview

OS Open Greenspace depicts the location and extent of spaces such as parks and sports facilities that are likely to be accessible to the public. Where appropriate, it also includes Access Points to show how people get into these sites. Its primary purpose is to enable members of the public to find and access greenspaces near them for exercise and recreation.

As an open data product, this will become part of the OS Open Data portfolio, which currently consists of a range of datasets, such as OS Vector Map District, OS Open Map - Local, OS Terrain 50 and OS Open Roads. For more information on the open data portfolio, see the [OS website](#).

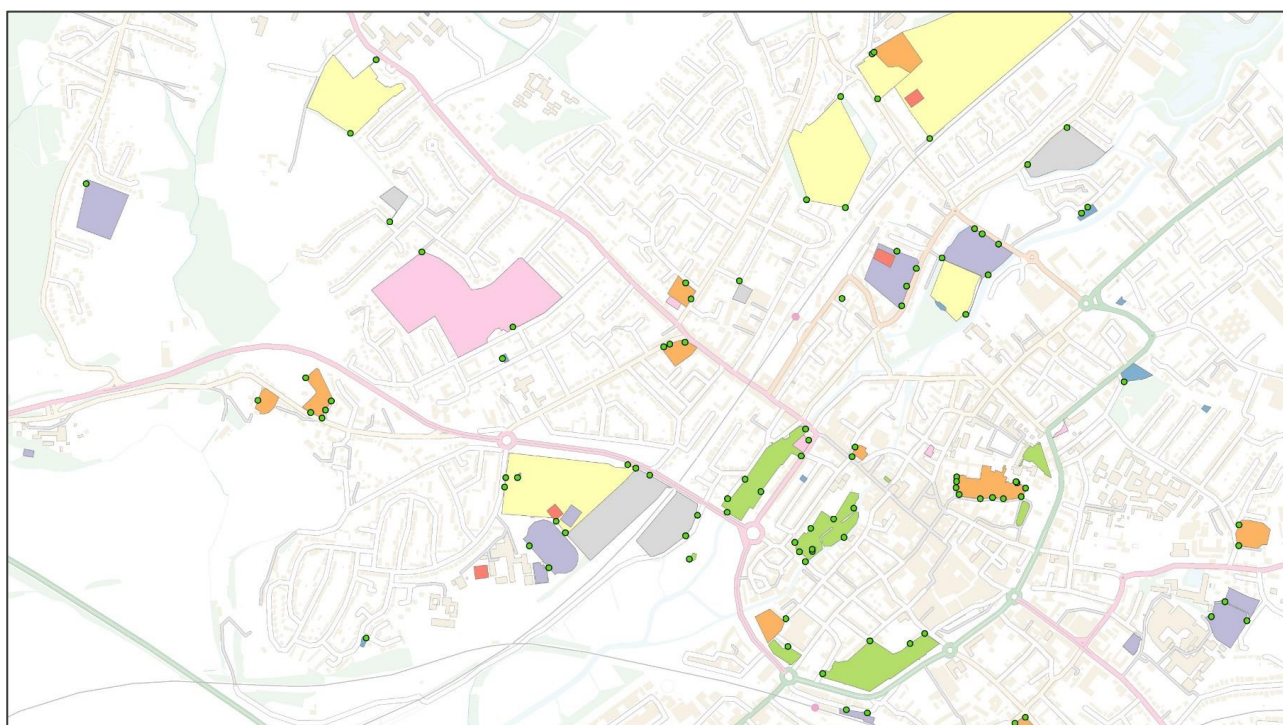


Figure 1: Open Greenspace data showing polygons and Access Points over OS Open Map – Local.

2. Data classification and structure

2.1 Data structure

The data structure is described below by means of a UML class diagram and accompanying tables containing text.

2.2 Open Greenspace classification scheme

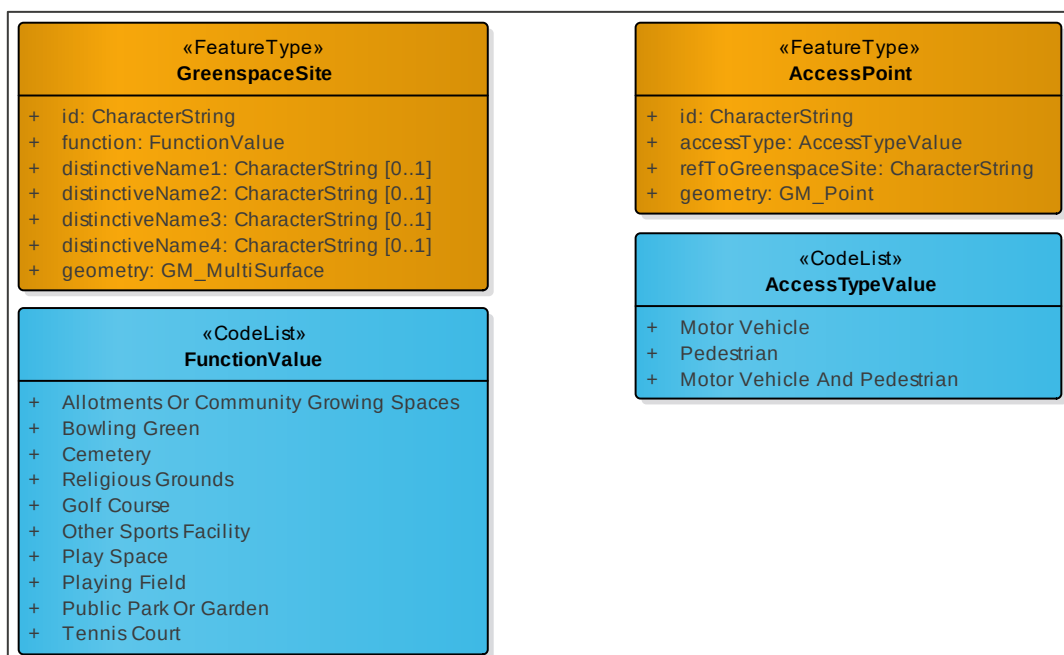
Open Greenspace consists of the following classifications:

- Public parks or gardens
- Play spaces
- Golf courses
- Sports areas or playing fields
- Churchyards or burial grounds
- Allotments or community growing spaces

A full table of these classifications and associated Ordnance Survey real-world terms and their definitions is included in the code lists below.

2.3 Model overview

Greenspace Open is constructed as per the following UML diagram.



2.4 Feature types

Open Greenspace contains two feature types: GreenspaceSites and AccessPoints.

2.4.1 GreenspaceSites

GreenspaceSites are polygons which depict the outer limits of a feature which may or may not have a physical boundary. These can be single or multi-part polygons.

Attributes

«FeatureType» GreenspaceSite		
Definition: A spatial area object describing the geometry, extent, and function of a real-world feature. This does not imply a physical boundary.		
Attribute: id		
Definition: Unique identifier of the site		
Type: CharacterString	Length: 38	Multiplicity: [1]
Attribute: function		
Definition: Description of the function of the site		
Type: FunctionValue	Length: 40	Multiplicity: [1]
Attribute: distinctiveName1		
Definition: The name of the site		
Type: CharacterString	Length: 254	Multiplicity: [0..1]
Attribute: distinctiveName2		
Definition: The name of the site		
Type: CharacterString	Length: 254	Multiplicity: [0..1]
Attribute: distinctiveName3		
Definition: The name of the site		
Type: CharacterString	Length: 254	Multiplicity: [0..1]
Attribute: distinctiveName4		
Definition: The name of the site		
Type: CharacterString	Length: 254	Multiplicity: [0..1]
Attribute: geometry		
Definition: The geometry of the greenspace area.		
Type: GM_MultiSurface	Length:	Multiplicity: [1]

Code List

Code List: FunctionValue http://www.os.uk/xml/codelists/OpenFunctionValue.xml Value defining the function of the greenspace	
Value	Description
Allotments or Community Growing Spaces	Areas of land for growing fruit, vegetables and other plants, either in individual allotments or as a community activity. Produce is for the grower's own consumption and not primarily for commercial activity.
Bowling Green	A specially prepared area intended for playing bowls.
Cemetery	Areas of land associated with burial areas.
Religious Grounds	Areas of land associated with churches and other places of worship. Only included where there are significant areas of greenspace (over 500m ² of natural space - identified as surfaces that are not manmade, such as grass, woodland and bare earth).
Golf Course	A specially prepared area intended for playing golf.
Other Sports Facility	Land used for sports not specifically described by other categories. Includes those facilities where participation in sport is the primary use of the area.
Play Space	A specially prepared area intended for children's play, usually linked to housing areas or parks and containing purpose-built equipment. Not captured if within schools or paid-for tourist attractions.
Playing Field	Large, flat areas of grass or specially designed surfaces, generally with marked pitches, used primarily for outdoor sports, i.e. football, rugby, cricket.
Public Park or Garden	Areas of land designed, constructed, managed and maintained as a public park or garden. These normally have a defined perimeter and free public access, and generally sit within or close to urban areas. Access is granted for a wide range of uses and not usually restricted to paths or tracks within the area. May include areas with managed facilities such as benches and flowerbeds, and more natural areas.
Tennis Court	A specially prepared area intended for playing tennis.

2.4.2 AccessPoint

Access Points are point geometries which depict either vehicular or pedestrian access into a Greenspace Site.

Attributes

«FeatureType» AccessPoint		
Definition: The spatial object type defining a point feature which would normally lie on the boundary of a site extent where there is access into or out of the site.		
Attribute: id		
Definition: The unique identifier of the Access Point.		
Type: CharacterString	Length: 38	Multiplicity: [1]
Attribute: accessType		
Definition: Describes the nature of the access permitted at the Access Point.		
Type: AccessTypeValue	Length: 40	Multiplicity: [1]
Attribute: refToGreenspaceSite		
Definition: The unique identifier of the greenspace site to which the Access Point relates.		
Type: CharacterString	Length: 38	Multiplicity: [1]
Attribute: geometry		
Definition: Location of the point		
Type: GM_Point	Length:	Multiplicity: [1]

Code List

Code List: AccessTypeValue	
http://www.os.uk/xml/codelists/AccessTypeValue.xml	
Value describing the type of access indicated by the Access Point	
Value	Description
Motor Vehicle	Access Point permits access to motor vehicles.
Motor Vehicle and Pedestrian	Access Point permits access to motor vehicles and pedestrians.
Pedestrian	Access Point permits access to pedestrians.

2.5 Precision

Open Greenspace geometry is published with a precision of two decimal places.

3. Formats

OS Open Greenspace is available in three formats: GML 3.2.1, ESRI Shapefile and OGC GeoPackage.

3.1 Attribute mapping

The naming of attributes between GML, GeoPackage and ESRI Shapefile will be different. For example, shapefiles are limited to 10 characters for an attribute name. Therefore, the following tables map the attribute names across the formats.

Uniquely among the formats, GML contains an attribute which describes the geometry of the feature; this is not applicable for the shapefile and GeoPackage formats as they apply their geometry visually.

3.1.1 GreenspaceSite

GML attribute	GeoPackage attribute *	ESRI Shapefile attribute
id	id	id
function	function	function
distinctiveName1	distinctiveName1	distName1
distinctiveName2	distinctiveName2	distName2
distinctiveName3	distinctiveName3	distName3
distinctiveName4	distinctiveName4	distName4

3.1.2 AccessPoint

GML attribute	GeoPackage attribute *	ESRI Shapefile attribute
id	id	id
accessType	accessType	accessType
refToGreenspaceSite	refToGreenspaceSite	refToGSite

* GeoPackage also comes with a 'fid' identifier. This is a procedurally generated, non-persistent number required for some GIS applications to be able to read the data.

3.2 GML

The Open Greenspace product is supplied in GML version 3.2.1. This section describes how Open Greenspace is defined in GML. An understanding of XML (eXtensible Mark-up Language) and XML schemas is required.

GML is an XML grammar for expressing geographic features. GML serves as a modelling language for geographic systems as well as an open interchange format for geographic transactions on the Internet. More information can be found on the Open Geospatial Consortium (OGC):
<http://www.opengeospatial.org/standards/gml>

The XML specifications that GML is based on are available from the World Wide Web Consortium (W3C): <http://www.w3.org>.

Information about Unicode and UTF-8, the character encoding we have chosen, is available on the Unicode Consortium website: <http://www.unicode.org>.

3.2.1 Use of examples

Any use of examples in this section that mention specific data content are to be taken as examples only.

3.2.2 Schema overview and location

XML schemas are used to define and validate the format and content of the GML. The GML v3.2.1 specification provides a set of schemas that define the GML feature constructs and geometric types. These are designed to be used as a basis for building application-specific schemas, which define the data content.

The schema document defines the namespace,
<http://namespaces.ordnancesurvey.co.uk/Open/Greenspace/1.0>; this is defined in the XSD at:
<https://www.ordnancesurvey.co.uk/xml/open/greenspace/1.0/OSOpenGreenspace.xsd>.

3.2.3 Code list dictionaries

- <http://www.os.uk/xml/codelists/OpenFunctionValue.xml>
- <http://www.os.uk/xml/codelists/AccessTypeValue.xml>

3.2.4 Example record

GreenspaceSite

```
<os:featureMember>
  <ogsp:GreenspaceSite gml:id="id45CEA121-A4AA-51CD-E053-2362A00A34C0">
    <ogsp:function
codeSpace="http://www.os.uk/xml/codelists/OpenFunctionValue">Playing Field</ogsp:function>
    <ogsp:distinctiveNameI>Northfield Playing Field</ogsp:distinctiveNameI>
    <ogsp:geometry>
      <gml:MultiSurface gml:id="id45CEA121-A4AA-51CD-E053-
2362A00A34C0-0" srsName="urn:ogc:def:crs:EPSG::27700" srsDimension="2">
        <gml:surfaceMember>
          <gml:Surface gml:id="id45CEA121-A4AA-51CD-E053-2362A00A34C0-1">
            <gml:patches>
              <gml:PolygonPatch>
                <gml:exterior>
                  <gml:LinearRing>
                    <gml:posList>391043.16
807998.57 391057.2 808010.9 391074.25 808015.9 391052 808085.2 391065.3 808092.15 391044.3
808164.2 390937.15 808226.55 390926.25 808254.55 390928.74 808288.15 390793.65 808280.8
390788.65 808305.8 390782.15 808316.1 390754.55 808328.6 390739.05 808327.05 390684 808228.65
390728.45 808181.85 390753.38 808138.18 390889.44 808144.4 390893.2 808096.18 390893.24
808095.64 390893.26 808095.52 390895.05 808072.4 390902.79 807954.2 390907.39 807954.45
390916.95 807961.9 391023.85 807965.95 391027.5 807985.05 391043.16 807998.57</gml:posList>
                  </gml:LinearRing>
                </gml:exterior>
              </gml:PolygonPatch>
            </gml:patches>
          </gml:Surface>
        </gml:surfaceMember>
      </gml:MultiSurface>
    </ogsp:geometry>
  </ogsp:GreenspaceSite>
</os:featureMember>
```

AccessPoint

```
<os:featureMember>
  <ogsp:AccessPoint gml:id="id8DE0ED21-B0F6-4F69-8008-CD2AD0B537CD">
    <ogsp:accessType
      codeSpace="http://www.os.uk/xml/codelists/AccessTypeValue">Motor Vehicle And
      Pedestrian</ogsp:accessType>
    <ogsp:refToGreenspaceSite>45CEA121-D8B7-51CD-E053-
      2362A00A34C0</ogsp:refToGreenspaceSite>
    <ogsp:geometry>
      <gml:Point gml:id="id8DE0ED21-B0F6-4F69-8008-CD2AD0B537CD-
        0" srsName="urn:ogc:def:crs:EPSG::27700" srsDimension="2">
        <gml:pos>394186.08 809827.7</gml:pos>
      </gml:Point>
    </ogsp:geometry>
  </ogsp:AccessPoint>
</os:featureMember>
```

3.3 ESRI Shapefile

OS Open Greenspace is supplied as an ESRI Shapefile. An ESRI Shapefile is an open file format to store geometry and attribute information about spatial features. It is a plug-in and play format, allowing you to drag and drop the files directly into your GIS application.

Example Record

GreenspaceSite

id	function	distName1	distName2	distName3	distName4
7367CB29-EDCE-4A5F-E053-A03BA40AD968	Cemetery				
7367CB01-3F18-4A5F-E053-A03BA40AD968	Cemetery				
7367CB01-3CBC-4A5F-E053-A03BA40AD968	Cemetery				
7367CB01-3B02-4A5F-E053-A03BA40AD968	Religious Grounds	St John's Kirk			

AccessPoint

id	accessType	refToGSite
002488DD-BB2E-4127-B26C-5DF2B2BC7ADE	Pedestrian	7367CB29-EDC0-4A5F-E053-A03BA40AD968
028E1421-8643-4A09-98D5-F020D3D0BB0B	Pedestrian	7367CB29-EDD5-4A5F-E053-A03BA40AD968
149B39CB-B512-4EA0-8510-3708CDDDB76CE	Pedestrian	7367CB01-3A96-4A5F-E053-A03BA40AD968
2507B2B4-50DF-4CE8-A20E-12491C261AB3	Motor Vehicle And Pedestrian	7367CB29-EDBC-4A5F-E053-A03BA40AD968

3.4 OGC GeoPackage

GeoPackage (*.gpkg) is an open, standards based, data format as is defined by the Open Geospatial Consortium (OGC). It is designed to be a lightweight format that can contain large amounts of varied and complex data in a single, easy to distribute and ready to use file.

GeoPackage offer users the following benefits:

- The single file is easy to transfer and offers the end-user a rich experience.
- Attribute names are not limited in length, making it user friendly.
- No file size limit, so lots of data can be easily accommodated.
- It supports raster, vector and database formats, making it a highly versatile solution.
- It is an OGC Standard.
- In most cases, it is a plug-in-and-play.

GeoPackage was released in 2014 and so is a relatively new format. As a result, some older software packages may have trouble loading it or may need a plugin in order to do so. If this is the case, your version of GIS may need updating. For example, QGIS software, as of version 2.18 (October 2016), is able to interact with GeoPackage files without needing additional plugins or settings. Earlier versions will either require a plugin or will not be able to interact with this format.

Example record

GreenspaceSite

fid	id	function	distinctiveName1	distinctiveName2	distinctiveName3	distinctiveName4
1	id81B3336A-A7C3-33F1-E053-A03BA40A2BEA	Cemetery				
2	id81B33342-0374-33F1-E053-A03BA40A2BEA	Cemetery				
3	id81B33342-0118-33F1-E053-A03BA40A2BEA	Cemetery				
4	id81B33341-FF60-33F1-E053-A03BA40A2BEA	Religious Grounds	St John's Kirk			
5	id81B332FE-CF2F-33F1-E053-A03BA40A2BEA	Play Space				

AccessPoint

fid	id	accessType	refToGreenspaceSite
1	idF85D59BE-9E70-40A9-AA62-1B271EE559C7	Pedestrian	id81B33342-0374-33F1-E053-A03BA40A2BEA
2	id8A4DB610-3170-400F-AEBD-B37C26B9D86B	Pedestrian	id81B33342-0118-33F1-E053-A03BA40A2BEA
3	idE89FDD9B-7470-400B-95B5-D6E2F5949E08	Motor Vehicle And Pedestrian	id81B33341-FF60-33F1-E053-A03BA40A2BEA
4	id57EDC198-A894-4F50-A218-C1672C4232CE	Pedestrian	id81B332FE-CF2F-33F1-E053-A03BA40A2BEA
5	idEF1C7874-E6C2-4EAB-96FF-30D10B1771D4	Motor Vehicle And Pedestrian	id81B33341-FEF4-33F1-E053-A03BA40A2BEA

3.5 Vector tiles

OS Open Greenspace is provided in MBTiles, which is an open specification tileset format used for vector tiles. This is designed to be simple and efficient to load as a single file, high resolution and customisable. The data is supplied in Web Mercator projection (ESPG:3857). Vector tiles schema is detailed in the following table.

3.5.1 Vector tiles schema

Layer	Feature types	Attributes	Zoom levels						
			0 - 8	9	10	11	12	13	14
greenspace_site	Allotments Or Community Growing Spaces	id, function, distinctive_name_1, distinctive_name_2, distinctive_name_3, distinctive_name_3							
	Bowling Green								
	Cemetery								
	Golf Course								
	Other Sports Facility								
	Play Space								
	Playing Field								
	Public Park Or Garden								
	Religious Grounds								
	Tennis Court								
access_point	Motor Vehicle	id, access_type							
	Motor Vehicle and Pedestrian								
	Pedestrian								

